

Information Visualization

Week 4: Visual Encoding & Perception

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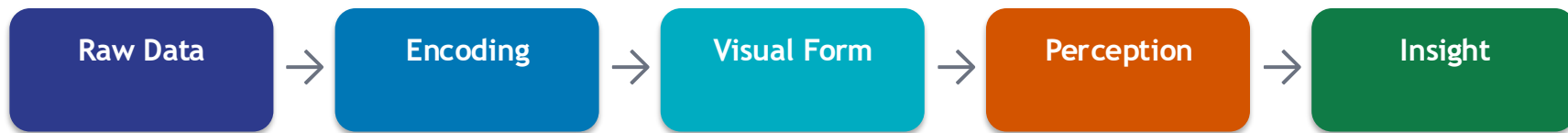
Learning Objectives

By the end of this lecture, you will be able to:

- 1 Understand the 'What' (data types) and 'How' (visual channels) of data visualization
- 2 Identify and classify marks (point, line, area) and channels (position, color, size, shape)
- 3 Apply the Expressiveness and Effectiveness principles to choose the right encoding
- 4 Recognize cognitive and perceptual constraints that affect how viewers read charts
- 5 Design visualizations that respect human perception and minimize misinterpretation

The Big Picture: Why Encoding Matters

Visualization is not decoration — it is a cognitive tool. The right encoding lets the human brain detect patterns, compare quantities, and identify outliers instantly.



Key Idea

Translating abstract data into actionable insights requires mapping data variables to visual properties that the human perceptual system can decode quickly and accurately.

Discussion: Think of a chart that instantly communicated an insight. What made it work?

What Is Visual Encoding?

Visual encoding is the process of mapping data variables to marks and channels. Every visualization you see is a set of encoding choices — some good, some bad.

Data Variables

Quantitative (e.g., temperature, salary)

Ordinal (e.g., low / medium / high)

Categorical (e.g., country, product type)



Visual Properties

Marks: point, line, area

Channels: position, color, size, shape, tilt

Composition: layouts, facets, layers

Note: The quality of your encoding determines how fast and accurately viewers interpret your data.

The Human in the Loop

Perception is the bottleneck of data communication. We process ~10 million bits/sec visually but only ~40 bits/sec consciously.

Eyes

Pre-attentive processing in <200ms: color, size, orientation pop out before we 'think.'

Brain

Working memory holds 7 ± 2 chunks. Overload = confusion. Legends > 7 items break down.

Bias

We see what we expect. Framing, context, and anchoring shape how we interpret data.

Implication: A visualization is only as good as the human's ability to perceive it.

Introduction to Marks

Marks are the basic geometric elements in a visualization. They represent items or nodes in your data.



Point

A dot representing a single data item.



Line

A stroke connecting two or more items.



Area

A filled 2-D region representing volume.

Example: On a map, cities = Point marks, roads = Line marks, country borders enclose Area marks. A scatterplot uses only Point marks; a line chart uses Line marks.

Exercise: For a stock price over 12 months, which mark type is most appropriate and why?

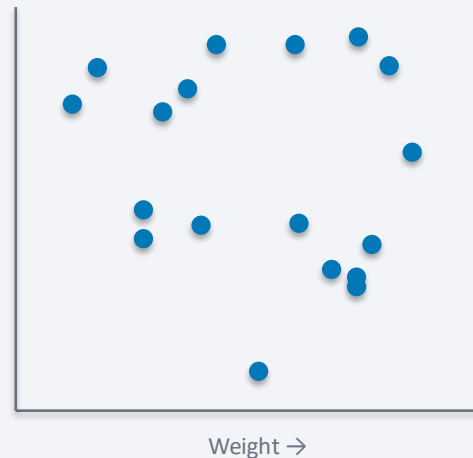
Point Marks: Representing Individual Items

Point marks represent individual data items. They are zero-dimensional (no inherent length or area) and rely entirely on channels like position, color, and size to convey information.

Common uses:

- Scatterplots — two quantitative variables
- Dot plots — distribution of one variable
- Geographic maps — locations as points
- Bubble charts — point + size channel

Scatterplot



Note: Point marks are the most versatile — they can carry color, size, and shape channels simultaneously.

Line Marks: Connections and Trends

Line marks represent connections or trends. They are 1-dimensional and imply continuity — joining two points with a line says 'these are related in sequence.'

Common uses:

- Line charts — trends over time
- Network/graph diagrams — relationships
- Slope charts — change between two points
- Parallel coordinate plots

Line Chart



Discussion: When is it misleading to connect points with a line? Hint: think about categorical x-axes.

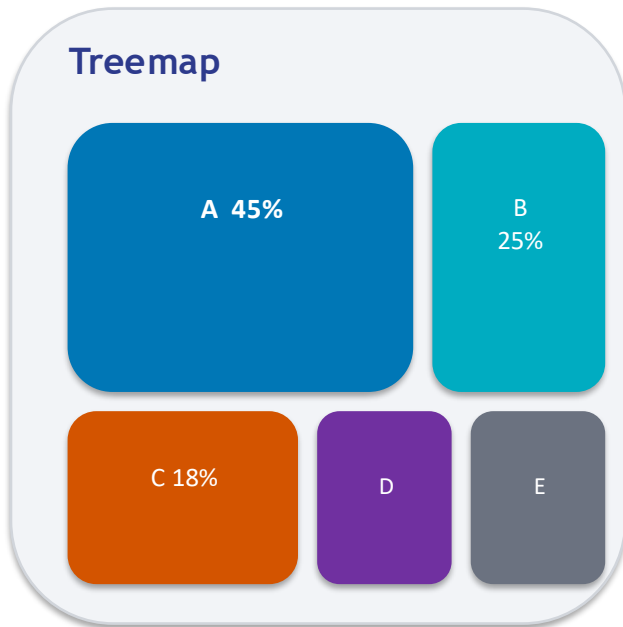
Area Marks: 2-D Regions and Volumes

Area marks represent 2-D filled regions. They encode data as the extent of a surface, which is useful for proportions, hierarchies, and geographic data.

Common uses:

- Choropleth maps — shaded geographic regions
- Treemaps — hierarchical proportions
- Heatmaps — density on a 2-D grid
- Stacked area charts — part-to-whole over time

Note: Area perception is less accurate than position or length (Stevens' Law, $n \approx 0.7$).



Introduction to Channels

Channels control the visual appearance of marks — they are how we encode the dimensions of our data.

Position

Where on the canvas (x, y)

Color

Hue, saturation, luminance

Size

Length, area, volume

Shape

Glyph / icon type

Tilt

Angle / orientation

Coming up: We will rank these channels by effectiveness for different data types.

Position Channels: The Most Powerful Encoding

Horizontal (x) and vertical (y) position is the most accurate channel. Our brains are evolved to judge distances on a plane with high precision.

Example: "The further right a dot is, the heavier the person."

Why Position Wins

We perceive spatial distance instantly and accurately

Position supports both comparison and pattern detection

Reserve position for your most important variable

Height vs. Weight



Color Channels: Identity vs. Magnitude

Hue → Categorical (Identity)

Different colors for different categories. Example: Each state a distinct color on a map.



Saturation / Luminance → Magnitude

Shades of one color for ordered data. Example: Choropleth — darker = higher density.



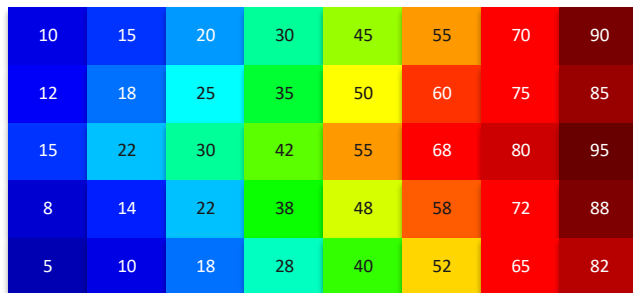
Pro Tip: Avoid Rainbow Scales

Rainbow color maps create false boundaries, are not perceptually uniform, and fail for color-blind users. Use sequential (light→dark) or diverging (two-hue) palettes instead.

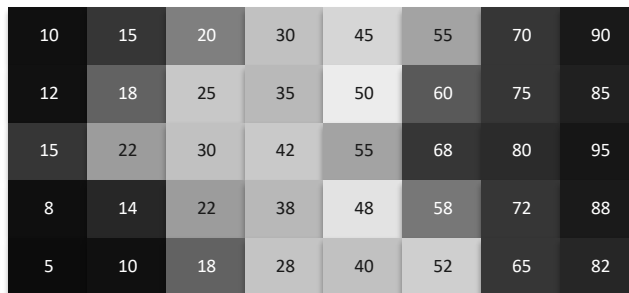
Exercise: Take a rainbow-colored heatmap. Convert it to greyscale. Can you still read the data?

The Greyscale Test on Rainbow Heatmaps

Rainbow Colormap (Jet)



→ Converted to Greyscale



✗ NO! Values 20 (cyan), 55 (yellow-green), and 80 (red) all map to similar grey tones. Green (val 30) appears BRIGHTER than yellow (val 55). The data ordering is destroyed.

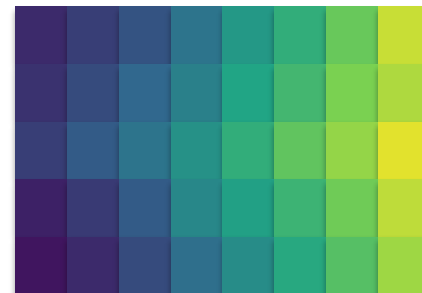
Why Does Rainbow Fail the Greyscale Test?

Non-monotonic luminance: Green has higher luminance than red or blue — greyscale mapping is scrambled.

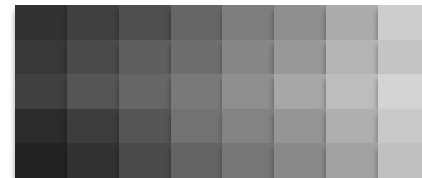
False boundaries: Sharp hue transitions (cyan→green, yellow→red) create perceived edges not in the data.

Fix: Use Viridis, Magma, or Inferno — luminance increases monotonically so greyscale always works.

✓ Fix: Viridis



→ Its greyscale:



✓ Monotonic luminance
= order preserved!

Shape Channels: Categorizing by Glyph

Shape is an identity channel — it tells us 'what kind' but cannot convey 'how much.' Different glyphs distinguish categories within point marks.



Circle



Diamond



Square



Triangle



Cross

Best Practices

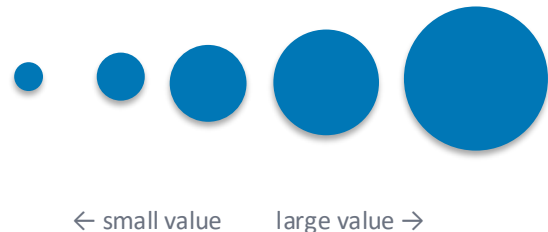
Limit to 5–7 distinct shapes. Beyond that, shapes become hard to discriminate. Combine shape + color for richer encoding (e.g., species in a scatterplot).

Size Channels: Length, Area, Volume

Size can encode quantitative data through length, area, or volume. Caution: Humans underestimate area growth (Stevens' Law, $n < 1$).

Be Careful with 3D

Volume is the hardest for humans to judge. A sphere with 2× radius has 8× the volume — viewers cannot perceive this correctly.



Recommendation: For bubble charts, scale by area (not radius) so that perceived size is proportional to data value. Always include a size legend.

Tilt and Angle Channels

Angle and orientation encode data via rotation. The most common example: pie chart slices. Angle is perceived less accurately than position or length — use with caution.

Recommendation

If you must use a pie chart, limit to 2–3 slices and label values directly. A horizontal bar chart encodes the same data more accurately using position and length.

Why Pie Charts Struggle

Humans compare angles poorly
Adjacent slices distort perception
3D and exploded pies are worse
5+ slices → unreadable
Bar chart almost always wins

Discussion: Can you think of a case where a pie chart is genuinely the best choice?

The Expressiveness Principle

"Tell the truth: show all the data, and only the data."

An encoding is expressive if it conveys exactly the information in the data — no more, no less. The visual structure should mirror the data structure.

Common Violations

Using ordered channels (saturation) for categorical data → implies false ranking

Using identity channels (shape) for quantitative data → can't perceive 'how much'

Truncating axes → hides data and exaggerates trends

Adding 3D effects → introduces perceptual distortion not present in the data

The Effectiveness Principle

Match the importance of the data to the 'strength' of the channel. Most important variable → strongest channel (position).

← Most accurate (strongest)

1. Position on common scale

2. Position on unaligned scale

3. Length

4. Tilt / Angle

5. Area / Color saturation

← Least accurate (weakest)

Note: This ranking comes from Cleveland & McGill's (1984) landmark perceptual experiments.

Stevens' Psychophysical Power Law

We don't perceive changes in a 1:1 linear fashion.

Formula: $S = I^n$ (S = sensation, I = intensity, n = exponent)

Length ($n \approx 1.0$)

Perceived accurately —
what you see matches data.

Area ($n \approx 0.7$)

Underestimated —
we see growth as less.

Brightness ($n > 1$)

Over-perceived —
small changes feel large.

Key Takeaway: Doubling the radius of a circle does not look like 'double' — it looks like 4× to a human!

Ranking Channels: Quantitative Data

Use this hierarchy when encoding numbers:

1

Position on common scale

Bar chart — compare on same axis

2

Position on unaligned scale

Multiple scatterplots side by side

3

Length

Be careful with baselines!

4

Tilt / Angle

Pie charts — hard to compare

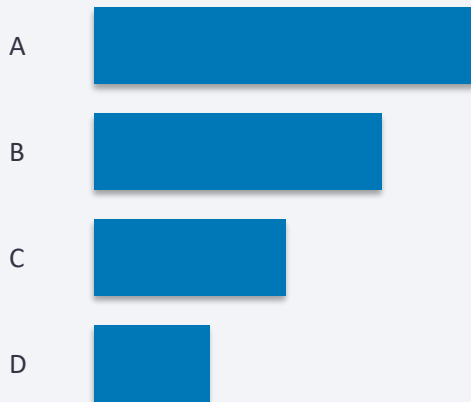
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Area

Bubble charts — worst precision

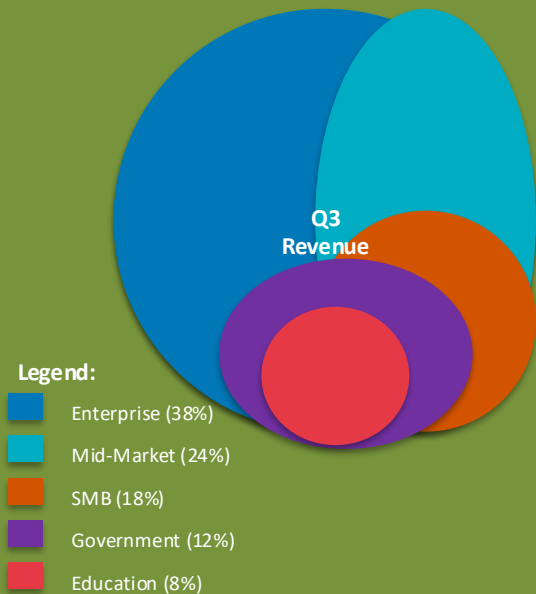
Excercise: Re-encode a pie chart as a bar chart and compare readability.

Same data, bar chart wins



Exercise Answer: Pie Chart vs. Bar Chart

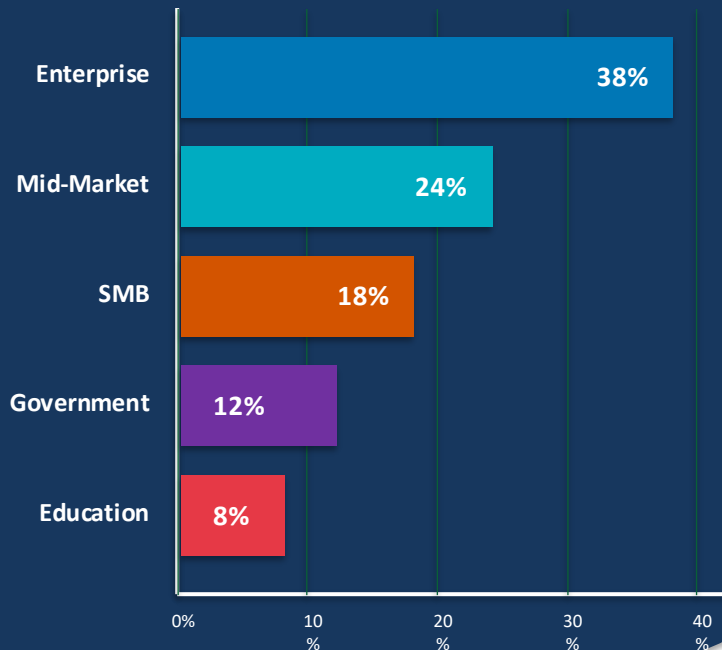
✗ Pie Chart (Angle Encoding)



Try to answer from the pie:

Is Mid-Market 2× or 3× Education? Is SMB closer to Gov or Mid-Market?

✓ Bar Chart (Position Encoding)



Now answer from the bar chart:

Mid-Market is 3× Education (24 vs 8). SMB is closer to Gov (18 vs 12). Instant!

Ranking Channels: Categorical Data

For 'what kind' (identity) data, use this hierarchy:

1. Spatial Grouping

Place related items near each other on screen

2. Color Hue

Assign distinct hues to categories

3. Shape

Use different glyphs (circle, triangle, square)

4. Motion / Animation

Flickering or moving elements draw attention

Note: Spatial grouping is the strongest because it leverages the Gestalt principle of proximity.

Magnitude vs. Identity Channels

Some channels encode 'how much' (magnitude), others encode 'what kind' (identity). Mixing them up violates expressiveness.

Magnitude: "How much?"

- Position on scale
- Length
- Area
- Color saturation / luminance
- Tilt / Angle

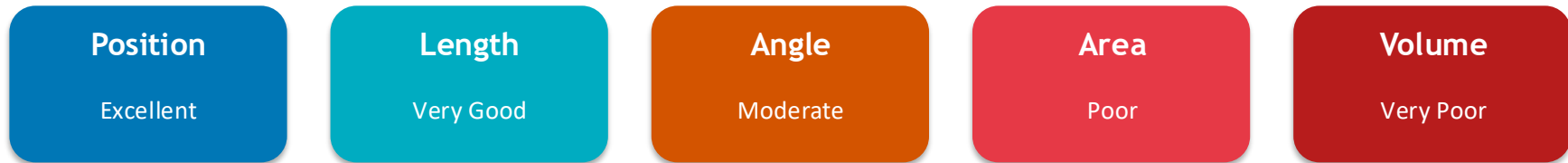
Identity: "What kind?"

- Spatial region
- Color hue
- Shape
- Motion
- Texture

Use magnitude channels for quantitative data; identity channels for categorical data.

Accuracy of Perception

Humans are great at judging position but terrible at judging area. Cleveland & McGill (1984) quantified this.



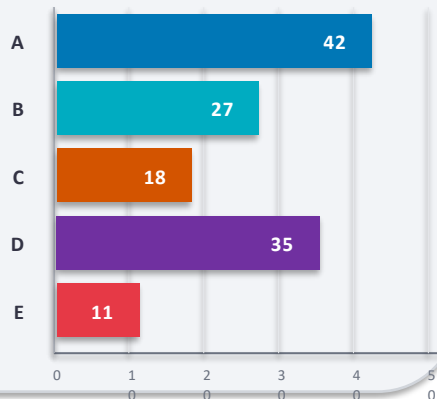
This is why bar charts (position) outperform pie charts (angle) and bubble charts (area) for the same data. When precision matters, always favor position encoding over weaker channels.

Exercise: Show the same 5 values as a bar chart, pie chart, and bubble chart. Ask a friend to estimate the values from each.

5 Actual values: A = 42 B = 27 C = 18 D = 35 E = 11 Task: Estimate each value without labels. Then compare accuracy.

✓ BEST

Bar Chart (Position)



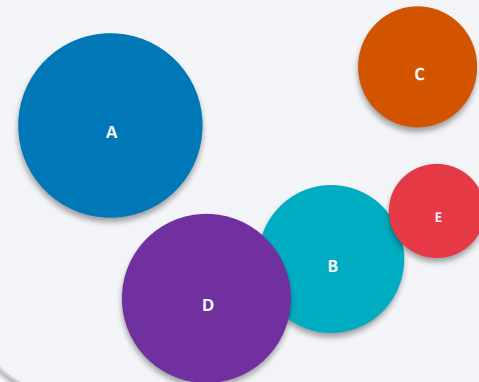
△ MODERATE

Pie Chart (Angle)



✗ WORST

Bubbles (Area)



Typical Estimation Results (based on perceptual research)

	Actual	Bar Est	Err	Pie Est	Err	Bub Est	Err
A	42	41	±1	36	±6	34	±8
B	27	27	±0	22	±5	20	±7
C	18	18	±0	15	±3	14	±4
D	35	34	±1	30	±5	28	±7
E	11	11	±0	10	±1	13	±2
Mean Abs. Error →			±0.6		±4.6		±6.0

Why the Accuracy Gap?

±0.6

Bar — Position

Stevens' $n \approx 1.0$. Position on a common scale gives near-perfect linear perception.

±4.6

Pie — Angle

Angles are hard to compare, especially non-adjacent slices. Plus: requires legend lookup.

±6.0

Bubble — Area

Stevens' $n \approx 0.7$. We underestimate large areas and overestimate small ones systematically.

Position > Angle > Area — Cleveland & McGill (1984)

Discriminability

Can the user actually tell the difference between these values?
The more distinct values you encode, the harder discrimination becomes.

Practical Limits

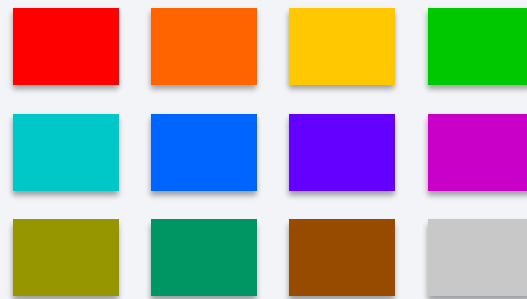
Color hue: ~6–8 distinguishable categories max

Shape: ~5–7 distinct glyphs max

Size: ~4–5 distinct sizes max

Beyond these limits, the encoding breaks down

Too Many Colors



Discussion: What strategies can you use when you have 15+ categories?

Separability vs. Integrality

When channels interfere with each other, viewers can't decode them independently.

✓ Separable: Position + Color

We can read x-position and color independently — they don't interfere.

Example: Scatterplot with colored points — we judge both location and category simultaneously.

✗ Integral: Width + Height

We perceive width $\times$ height as a single 'area' — hard to separate.

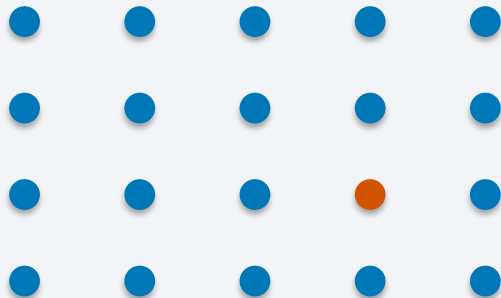
Example: Rectangles varying in both dims — viewers see 'big/small,' not individual dimensions.

Rule: Pair separable channels. Avoid encoding different variables on integral channels.

Pre-attentive Pop-out Effects

Pre-attentive features let users spot targets in < 200ms — before conscious thought kicks in.

Find the orange dot:



Pre-attentive Features

- Color hue (most powerful)
- Size (large among small)
- Orientation (tilted among straight)
- Shape (unique in a field)
- Motion (flickering element)

Design tip: Use pop-out strategically to draw the eye to the most important data point or anomaly.

Cognitive Load Theory

Don't make the user think harder than they have to. Every visualization imposes cognitive load.

Intrinsic Load

Complexity inherent in the data itself.

Extraneous Load

Bad design that adds unnecessary effort.

Germane Load

Effort that builds understanding & insight.

Goal: Minimize extraneous load (chartjunk, poor labels, confusing encodings). Maximize germane load (clear patterns, intuitive encodings, guided annotations).

Working Memory Limits: Miller's Law

The "Magic Number 7 (± 2)" — George Miller, 1956

Humans hold approximately 5–9 chunks of information in working memory at once. Legends with > 7 items overload memory — viewers re-check the legend repeatedly. Dashboards with 10+ separate charts force constant context-switching.



Design rule: If your legend has > 7 entries, simplify or use direct labeling. Direct annotation eliminates the memory lookup entirely.

Attention as a Limited Resource: Avoiding "Chartjunk"

X Chartjunk

- Decorative 3D effects
- Unnecessary gridlines
- Background images / patterns
- Gradient fills
- Redundant legends
- Clip art and stock photos

Each element competes for attention.

✓ Clean Design

- Minimal gridlines (or none)
- Direct labeling
- White space for breathing room
- Purposeful color (not decoration)
- Title that states the insight
- Every pixel earns its place

Attention is spent on the data.

Reading: Tufte, E. (1983) The Visual Display of Quantitative Information, Chapter 5.

Change Blindness

When a display changes (new slide, filter applied, axis rescaled), viewers often fail to notice differences — even large ones. Visual working memory is extremely limited.

Mitigation Strategies

Animate transitions so the eye can track what moved

Highlight changes explicitly (color the difference, add annotations)

Use small multiples side-by-side instead of flipping between views

Keep axes and scales consistent across comparisons

Discussion: Why do animated transitions in Tableau or D3 help reduce change blindness?

Mental Models

Match the chart to the user's expectations. Violating conventions forces costly mental model rebuilding.

Common Conventions (Western Audiences)

Time flows left → right

Up = more, Down = less

Red = danger/negative, Green = positive

Darker = higher density or intensity

Larger = bigger value

Example: Inverting the y-axis (higher numbers at bottom) can make a downward trend look like growth.

Context and Framing

Titles and annotations are editorial choices that guide interpretation — they are never neutral.

Title A: "Sales Are Declining"

Negative framing. The viewer focuses on the downward end and feels alarmed. Annotations highlight the decline.

Title B: "Sales Grew 200% Since 2019"

Positive framing. The viewer focuses on overall growth and feels optimistic. Annotations highlight the gain.

Exercise: Take one chart and write two titles — one alarmist, one optimistic. How does it change perception?

Decision Making Biases in Visualization

Visualizations can accidentally mislead due to cognitive biases inherent in human judgment.

Anchoring

The first number viewers see becomes their reference point.

Confirmation

Viewers see patterns that confirm existing beliefs.

Truncation

Starting y-axis at non-zero exaggerates small differences.

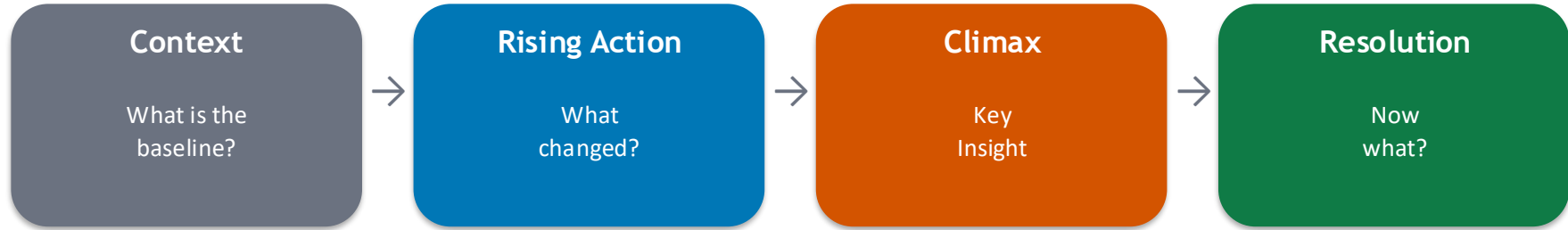
Framing

Percentages vs. raw counts tell different stories.

Note: Being aware of these biases is the first step to designing honest visualizations.

Narrative Flow in Data Storytelling

A good data narrative follows a story arc — structure your visualizations to lead to a conclusion.



1. Set context (baseline) → 2. Build tension (what changed) → 3. Reveal insight (so what?) → 4. Call to action (now what?). This structure works for reports, dashboards, and presentations.

Data-Ink Ratio (Edward Tufte)

Maximize the share of ink used to display actual data.

Data-Ink Ratio = Ink used for data / Total ink in the graphic

Strive for a ratio close to 1.0: remove decorative borders, background patterns, redundant gridlines. Erase non-data pixels: drop shadows, 3D effects, gradient fills. Simplify every element that doesn't contribute to understanding.

But don't over-minimize! Titles, axis labels, and annotations are essential for comprehension. The goal is not zero decoration — it is zero distraction.

Exercise: Take one of your charts and remove one element at a time. At what point does comprehension start to suffer?

Interactivity and Cognition

How Hover, Filter, and Zoom reduce cognitive burden.

Hover / Tooltip

Details on demand —
show data only when
the user asks.

Filter

Hide irrelevant data
so the user focuses
on what matters.

Zoom

Overview first, then
zoom into details.
(Shneiderman's mantra)

Brush / Link

Select in one view,
highlight in all —
cross-filtering.

"Overview first, zoom and filter, then details on demand." — Ben Shneiderman (1996)

Gestalt Principles of Perception

The brain automatically organizes visual information into meaningful patterns using these rules.

Proximity

Near = grouped

Similarity

Same look = grouped

Enclosure

Bounded = grouped

Continuity

Aligned = connected

Closure

Complete gaps mentally

Connection

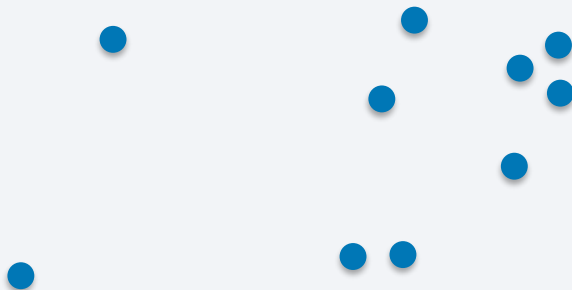
Lines link elements

These principles operate pre-attentively — you can't 'unsee' them. Use them, don't fight them.

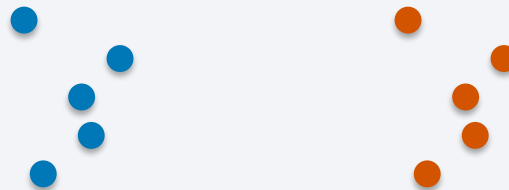
Gestalt: Proximity

Things that are close together are perceived as belonging to the same group.

Random — no grouping



Clustered — two groups seen



Application: In dashboards, place related metrics (Sales & Profit) in the same corner so the user links them mentally.

Gestalt: Similarity

Things that look alike are perceived as belonging to the same group — even if spatially separated.

When marks share a visual property (color, shape, size), we instinctively group them. In visualization: same color = same category across all charts. Be consistent — if 'Revenue' is blue in Chart 1, keep it blue in Chart 2.

Consistent palette:



Group A (blue)



Group B (orange)

Discussion: What happens when two dashboards use different colors for the same metric?

Gestalt: Enclosure & Continuity

Enclosure

Drawing a border around elements groups them visually.

Use in dashboards: panel borders, shaded backgrounds, card layouts to group related charts.

Even a subtle background shade creates strong grouping.

Continuity

Elements on a smooth path are perceived as connected.

Use in line charts: the connecting line implies a continuous trend.

Also applies to flow diagrams and process visualizations.

Note: These principles apply to dashboard layout, not just individual charts.

Contrast and Backgrounds

Ensuring the data stands out. Contrast is the most fundamental perceptual principle for readability.

Contrast Guidelines

Foreground vs. background: Data elements must have sufficient luminance contrast

Signal vs. noise: Gridlines and labels should be lighter/thinner than data marks

Highlight vs. context: Use saturation to make key data 'pop' while muting context

WCAG recommends minimum 4.5:1 contrast ratio for text, 3:1 for large elements

Common mistake: Dark or busy backgrounds competing with the data

Exercise: Take a chart with a colored background. Remove it. Does readability improve?

Visual Hierarchies

Using size, weight, and color to direct the viewer's eye. A visual hierarchy tells the viewer: 'Read this first, then this, then this.'

Create hierarchy with size (larger attracts first), weight (bold draws attention), color (saturated pops), and position (top-left is read first).



Apply to dashboards: the headline KPI should be the largest, boldest, most colorful element.

Optical Illusions in Visualization

How gridlines, shadows, and surroundings can distort data perception.

Mach Bands

Adjacent regions create illusory dark/light edges.
→ Avoid hard edges in sequential color maps.

Hermann Grid

Grey dots appear at gridline intersections.
→ Use light gridlines or remove them.

Contrast Effect

Same color looks different on different backgrounds.
→ Use consistent, neutral backgrounds.

Reading: Ware (2021) Information Visualization, Chapter 4: Color, Quantity, and Perception.

3D Visualization Hazards

The occlusion problem and perspective distortion make 3D charts dangerous.

Occlusion

Front objects hide rear objects — data becomes literally invisible.

Perspective Distortion

Farther elements appear smaller, biasing size judgments.

Depth Ambiguity

Without stereo vision, we cannot judge depth on a 2D screen.

Rule: Avoid 3D charts unless you are encoding a true third spatial dimension. 3D bar, pie, and scatter rotations almost always harm more than they help.

Putting It All Together: A Checklist

Use this checklist before finalizing any visualization:

- 1 Identify data types (quantitative, ordinal, categorical)
- 2 Choose marks (point, line, area) appropriate to the data
- 3 Map the most important variable to position (strongest channel)
- 4 Check expressiveness: does the encoding match the data type?
- 5 Check effectiveness: are channels ranked by importance?
- 6 Limit legend to ≤ 7 items; prefer direct labeling
- 7 Test for color blindness (greyscale test, blue/orange palette)
- 8 Remove chartjunk; maximize data-ink ratio
- 9 Write a title that states the insight, not just the topic
- 10 Get feedback from someone who hasn't seen the data

Case Study: Before vs. After

Re-encoding a 'bad' chart using the principles from this lecture.

✗ Poor Choices

- 3D exploding pie chart
- 12 categories (too many slices)
- Rainbow color scale
- No direct labels — requires legend
- Decorative clip art in background
- Title: "Q3 Revenue Breakdown"

✓ Effective Choices

- Horizontal bar chart (position)
- Top 5 categories + 'Other'
- Single-hue sequential palette
- Direct labels on each bar
- Clean white background
- Title: "Enterprise is 45% of Q3 revenue"

Homework: Find a bad chart online, apply 3 principles, and create a redesigned version.

Mini-Project 2 Kickoff: Interactive Dashboard

Introduction to the new requirements.

Requirements

1. Build an interactive dashboard (Tableau, D3, Plotly)
2. At least 3 coordinated views (linked brushing)
3. Encode ≥ 4 data dimensions via different channels
4. Apply Gestalt principles to organize layout
5. Test with 2 peers; report usability findings

Timeline

Week 5: Proposal + sketches
Week 7: Working prototype
Week 8: Final + peer review

More details in the course portal.

Key Takeaways

1. Marks (point, line, area) + Channels (position, color, size, shape) = visual encoding
2. Expressiveness: match encoding to data type — don't lie with your chart
3. Effectiveness: strongest channel (position) for most important variable
4. Cognitive limits: 7 ± 2 chunks, minimize extraneous load, direct-label when possible
5. Perception: leverage Gestalt, test for color blindness, avoid 3D unless essential

Ware (2021) Information Visualization • Munzner (2014) Visualization Analysis & Design • Tufte (1983) Visual Display • Cleveland & McGill (1984) Graphical Perception • Few (2012) Show Me the Numbers • Shneiderman (1996) The Eyes Have It